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BlueRev

Bio-based revitalisation
of local communities

KPIs and tools for selecting good practice methods

Deliverable 3.3

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the European Union

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Table of Abbreviations and Acronyms

Abbreviation	Meaning
CATWOE	Customers, Actors, Transformation, Worldview, Owners, and Environmental constraints
CCC	Context and Critical Context analysis
CFP	Common Fisheries Policy
CSC	Case Study Coordinators
DFBG	District of Fisheries and Blue Growth, Sicily
EMFAF	European Maritime, Fisheries and Aquaculture Fund
FBCD	Food & Bio Cluster Denmark
KPIs	Key Performance Indicators
LCA	Life Cycle Analysis
MCDA	Multi-Criteria Decision Analysis
MFA	Material Flow Analysis
NIBIO	The Norwegian Institute of Bioeconomy Research
R&D	Research and Development
RDT	Responsible Design Thinking
RISE	Research Institutes of Sweden
SROI	Social Return of Investment
UNIPA	University of Palermo
WP	Work Package
UiA	University of Agder

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1 Executive Summary

This deliverable consolidates BlueRev's conceptual work and field evidence into a practical, decision-ready framework for selecting and validating "good-practice" methods that reduce environmental footprints and maximise social value in the blue bioeconomy. Centred on a four-pillar set of Key Performance Indicators—environmental, social innovation, governance and business model—the report defines comparable metrics, maps them to fit-for-purpose tools, and shows how they are applied in the project's pilot regions in Denmark/Greenland, Estonia and Italy. The aim is to move beyond fragmented assessments to an integrated view of technical performance, social legitimacy, policy fit and economic viability.

On the environmental side, the framework operationalises life-cycle thinking through ISO-compliant LCA, Material Flow Analysis, energy intensity and water footprint, allowing teams to detect process hotspots, quantify circularity outcomes and align results with EU climate, circular-economy and water objectives. Social innovation is positioned as a performance domain in its own right, not a "soft" add-on, with Stakeholder Participation and Co-creation Depth indices, capacity-building metrics and Social Return on Investment making inclusiveness and community value auditable. Governance is treated as an active enabler of scale rather than a backdrop: Regulatory Adaptation, Policy Coherence and Network Density, complemented by perception-based indicators and the number of policy solutions proposed, provide a tractable way to evaluate whether institutions can host and expand new methods. Business-model viability is addressed through Responsible Design Thinking and the Sustainable Business Model Canvas, with indicators for problem–solution fit, iteration frequency, time-to-prototype, circular resource use, footprint reduction, sustainable revenue share, community value, stakeholder diversity, idea adoption and actionable outcomes, ensuring that co-creation remains evidence-based and responsive.

The toolkit spans participatory, analytical and digital methods—semi-structured interviews, focus groups and living-lab workshops; system mapping and governance/innovation-system analysis; LCA, MFA and SROI; blockchain traceability and GIS; and multi-criteria decision analysis (AHP, PROMETHEE) to integrate cross-pillar evidence into transparent rankings with sensitivity analysis. A concise guidance pathway helps teams choose methods that match data availability, stakeholder capacity and decision needs, with results feeding the WP5 dashboard and informing WP6 Policy Labs to close the loop between analysis, deliberation and action.

Early applications across the pilots indicate that the approach is transferable to diverse valorisation pathways—from fish by-products destined for health ingredients or pet food to algae-based innovations—while preserving comparability and auditability under FAIR data practices. It is suggested that MCDA is compulsory for consequential decisions, across sites KPI pack minimums are adopted, traceability and data governance are institutionalised from the beginning, and the resources available for facilitation are substantive rather than symbolic in terms of supporting social innovation processes. It also acknowledges limitations—data heterogeneity, uneven institutional capacity and the need for time-series to track some business-model indicators—and proposes next steps to harden targets, standardise protocols and publish open templates for recurring analyses. Taken together, the framework, toolchain and guidance provide a portable performance system that strengthens the legitimacy, scalability and durability of blue-bioeconomy innovations beyond the lifetime of the grant.

2 Introduction

2.1 Context & Rationale

BlueRev positions “good-practice” selection in the blue bioeconomy as a systems problem rather than a narrow optimisation exercise. The project responds to three intertwined needs observed across our pilot regions, Denmark/Greenland, Estonia and Italy. First, technical choices around side-stream valorisation must be assessed on comparable environmental grounds using life-cycle thinking so that circularity gains and hotspot processes are made visible and auditable. Second, method uptake depends on social legitimacy and co-ownership; inclusive innovation processes and the quality of co-creation shape adoption at least as much as engineering performance. Third, scale hinges on governance readiness and business-model viability, which are frequently the missing links between promising pilots and durable change. To address these gaps, the deliverable consolidates BlueRev’s conceptual work and field evidence into a decision architecture that integrates environmental metrics with social innovation, governance and sustainable business-model dimensions, and aligns evaluation practice with EU strategies on climate neutrality, circular economy, traceability and adaptive governance.

2.2 Scope

This deliverable defines the established project’s three-pillar KPI framework following the BlueRev structure, i.e. social innovation, governance and business model, to which a fourth is added to address the environmental aspects. This framework is coupled with a curated toolkit of participatory, analytical and digital methods that can be applied consistently across pilots while remaining sensitive to local context. The scope runs from indicator definition and operationalisation to method selection and early application in real valorisation pathways, with results designed to flow into decision support and deliberation spaces. Boundaries are explicit: environmental assessment follows ISO-compliant LCA with cradle-to-gate system limits and is complemented by MFA-based circularity, energy intensity and water footprint; social innovation is quantified through participation, co-creation depth, capacity building and SROI; governance focuses on regulatory adaptation, policy coherence, network density and problem-solving capacity; business-model viability is developed through Responsible Design Thinking and the Sustainable Business Model Canvas, tracked by adoption- and outcome-oriented indicators. The document is therefore both a reference for comparable measurement and a practical guide for selecting and justifying

methods in each pilot, with open-science and FAIR data practices embedded to support reuse and scale. Objectives

The main goals of task 3.3 are to:

Define KPI Framework: establish KPI categories and specific metrics for environmental footprint and enabling conditions (social, governance, economic).

Catalogue Analytical & Participatory Tools: identify tools to measure or select methods against these KPIs (e.g., LCA, workshops, SBMC, CLDs).

Provide Guidance for Application: illustrate how these KPIs and tools can be applied in the three pilot contexts.

The objective of D3.3 is to turn BlueRev's concepts and field insights (described in D3.1[1] and D3.2[2]) into a portable, decision-ready system that improves how "good practice" is identified, compared and defended. Concretely, the report establishes a common KPI language across the four pillars and translates it into measurable indicators that connect directly to pilot realities. It assembles a fit-for-purpose toolchain—from interviews, focus groups and Living Labs to LCA, MFA, SROI, GIS/blockchain and MCDA—that enables teams to gather robust evidence and integrate it transparently in rankings and trade-offs. It then provides application guidance so that regions with different data readiness and institutional capacity can still run credible evaluations and iterate as evidence accumulates. Finally, it demonstrates early use in the three pilot contexts to prove transferability, embeds outputs in the WP5 dashboard and WP6 Policy Labs for adaptive governance, and lays out the conditions for replication beyond the grant through standardised protocols, FAIR data management and traceable decision processes.

This document is organized as follows:

- Chapter 3 presents the methodology followed to identify, define and prioritise the KPIs and tools best suited for selecting good-practice methods in the BlueRev context
- Chapter 4 describes the four-pillar KPI framework, adapted from the BlueRev conceptual model.
- Chapter 5 gives the set of metrics to capture the most critical dimensions of performance, to enable the quantification and comparison of adopted methods.

[1] 10.5281/zenodo.7495021.
[2] 10.5281/zenodo.10591239.

- Chapter 6 presents the set of approaches to generate reliable, actionable data against chosen KPI; for each tool type, purpose, typical steps, and the outputs produced are described.
- Chapter 7 guides project teams to design a streamlined, context-sensitive evaluation workflow that leverages the strengths of each tool, avoiding unnecessary complexity.
- Chapter 8 illustrates how BlueRev hubs can deploy the metrics and tools outlined to their unique side-stream valorisation challenges.
- Chapter 9 highlights some important recommendations as a result of BlueRev experience, encouraging to move beyond traditional stakeholder engagement to fully activated social processes that solidify legitimacy, inclusivity, and sustainability in the world of innovation.
- Chapter 10 provides the conclusion for the work performed in BlueRev, highlighting how BlueRev pilots not only tested local solutions but also generated transferable knowledge that can guide Europe's coastal regions in their transition.

3 Methodology

This chapter outlines the step-by-step approach we followed to identify, define and prioritise the KPIs and tools best suited for selecting good-practice methods in the BlueRev context. We begin by reviewing foundational theory and previous project deliverables, then describe our protocol for gathering rich, on-the-ground insights, and finally explain how we synthesized qualitative and quantitative data into a cohesive framework. Together, these methods ensure that our KPI definitions and tool recommendations are both academically robust and practically relevant across all pilot regions.

The key selection pathways has been built on the literature and the other BlueRev deliverables review including the field protocol; starting from an initial longlist of candidate indicators, we achieved a concise and usable KPI set through five steps:

1. First, we compiled a long list. We extracted candidate indicators from the targeted literature review and key BlueRev outputs (incl. D3.1 and WP4 deliverables), and tagged each candidate to one of the four pillars (environmental, social innovation, governance, business model).

2. Then we assessed the field relevance and feasibility. Using the common protocol reported in D3.2, we triangulated longlist items against stakeholder evidence from interviews, focus groups and living-lab workshops to understand what can be credibly observed, measured, and influenced in real valorisation pathways. This was done across the four cases

3. After the assessment we applied narrowing criteria. Candidate indicators were retained only if they met all of the following: (i) decision relevance for “good practice” selection, (ii) measurability with available/collectable data, (iii) comparability across pilots (with defined units/scales), (iv) actionability (interpretable for changing practice), and (v) non-redundancy with other KPIs.

4. Following the narrowing we prioritised the list into a minimum “core KPI pack” and optional context KPIs. To balance comparability and local fit, we defined a cross-site minimum pack (mandatory for MCDA-based comparisons) and a small set of optional KPIs that regions can activate depending on data readiness and decision needs (e.g., traceability requirements, time-series availability).

5. Finally we operationalised the KPIs list. For each retained KPI we specified: definition, unit/scale, method/tool, data source, collection frequency, and interpretation rules (see measurement guidance below), enabling consistent

reporting and defensible trade-offs in WP5 dashboards and WP6 deliberation spaces.

3.1 Literature & Deliverable Review

To ground our work in established theory and project experience, we started with a targeted review of academic literature on social innovation, bioeconomy business models and governance structures, alongside key BlueRev outputs (Deliverables 3.1, 4.1–4.3). This stage allowed us to extract core concepts, typologies and existing KPI proposals, ensuring that our subsequent indicators and tools build directly on what has already been conceptualized and tested within the project.

3.2 Data Collection Protocol

Having established the theoretical base, we designed and implemented a qualitative data-gathering protocol (as detailed in Deliverable 3.2) to capture stakeholders' lived experiences with emerging methods. Through semi-structured interviews, focus groups and living-lab workshops, we elicited insights on feasibility, acceptance and operational challenges. This protocol provided the raw material for understanding how candidate methods actually perform against draft KPI criteria in the field. D3.2

3.3 Analytical Synthesis

In this final methodological phase, we brought together literature-derived concepts and field data through a thematic analysis. We mapped recurring themes from interviews and workshops onto KPI categories and then aligned each analytical or participatory tool to the specific measurement needs those themes revealed.

To move from an initial "universe" of possible indicators to a concise, decision-ready KPI pack, we followed a transparent narrowing process grounded in both evidence and feasibility. First, we compiled a longlist of candidate indicators from three sources:

(i) the targeted literature and standards underpinning the toolbox (e.g., LCA/ISO logic, MFA, participatory evaluation, SROI and MCDA),

(ii) the BlueRev conceptual work (i.e. D3.1) and other relevant project outputs, and

(iii) the empirical insights captured through the common data collection protocol (D3.2), ensuring that the indicator pool reflected what actually matters in the pilots and what can realistically be observed across hubs.

This resulted in an initial longlist of [N] candidate indicators across the four pillars.

We then reduced the longlist to a shortlist and final set through iterative screening and validation with the WP3 team and the case study coordinators/pilot teams. Indicators were retained only when they were:

(a) decision-relevant for selecting “good practice” methods,

(b) measurable with available or realistically collectable data,

(c) comparable across pilots via a clear unit/scale,

(d) actionable (i.e., interpretable for improving practice rather than merely describing context), and

(e) non-redundant (overlapping indicators were merged or removed).

Where indicators were valuable but not consistently measurable across sites (e.g., requiring long time-series or site-specific baselines), they were either reformulated into simpler proxies or treated as context KPIs rather than part of the cross-site minimum pack. Overall, the process moved from 49 longlist indicators → 33 shortlisted indicators → 26 final KPIs (of which 16 are “core” for cross-pilot comparison and 10 are optional/context-dependent, Figure 1). For applicability, each retained KPI is accompanied by practical measurement cues (definition, unit/scale, data source, frequency, aggregation rule, directionality/interpretation), enabling consistent assessment and transparent use in MCDA-supported comparisons.

The result is a clear tracing from high-level theory through on-the-ground realities to the practical metrics and instruments presented in subsequent chapters.

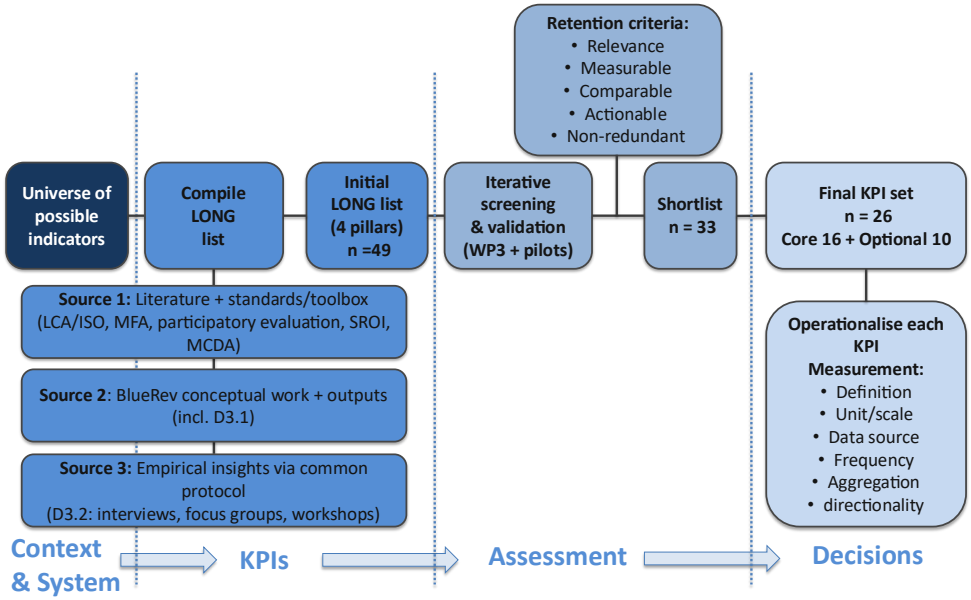


Figure 1: KPI selection pathway: from longlist to decision-ready KPI pack

3.4 Guideline for KPIs identification and application

Based on figure 1 we provide here a clear step-by-step guideline for users seeking to apply the BlueRev approach. Following this process interested stakeholders will be able to build their own relevant set of context-specific KPIs:

Step 1 — Define the decision context

Clearly specify *what* is being assessed and *why*: the valorisation pathway (e.g. marine side-stream use), the development stage (concept, pilot, scale-up), and the decision to support (comparison, selection, or improvement).

Step 2 — Be aware of the system

Frame the assessment using a system approach. In BlueRev we used four dimensions: **environmental**, **social innovation**, **governance**, and **business model**. This ensures systemic coverage and prevents single-dimension optimisation.

Step 3 — Build a longlist of candidate KPIs

Identify a broad set of potential indicators relevant to the pathway, drawing on life-cycle thinking, stakeholder engagement, governance readiness, and economic viability. At this stage, inclusiveness is prioritised over precision.

***Step 4 — Select a KPI set** (core and optional)*

Screen and prioritise indicators using explicit criteria: *decision relevance, data availability, measurability, comparability, actionability, and non-redundancy*.

This results in a small core KPI pack suitable for comparison, complemented by optional context-specific indicators.

Step 5 — Operationalise each KPI

For every selected KPI, define a simple measurement card (see Section 5) including: definition, unit or scale, data source, collection frequency, and interpretation rules (e.g. thresholds or direction of improvement).

Step 6 — Assess and interpret results transparently

Integrate the KPIs into a transparent assessment process, using e.g. multi-criteria decision analysis, to make trade-offs across environmental, social, governance, and economic dimensions explicit and traceable.

Following these steps allows external users to move from broad sustainability ambitions to a defensible, evidence-based KPI system that supports learning, comparison, and adaptive decision-making beyond the BlueRev project.

Table 1: KPI step by step selection

Step	Purpose	Key Questions	Output
1. Define the decision context	Clarify what the KPI system is meant to support	What valorisation pathway is assessed? At which development stage? What decision must be informed?	Clear assessment scope and decision objective
2. Look at the whole system	Ensure systemic and balanced assessment	Are different relevant dimensions addressed? E.g. environmental, social innovation, governance, and business model?	Structured assessment framework
3. Build a longlist of candidate KPIs	Capture all relevant	Which indicators reflect impacts, readiness, and	Broad KPI longlist

	performance aspects	viability across the four pillars?	
4. Select core and optional KPI	Focus on decision-relevant indicators	Which KPIs are currently measurable, comparable, non-redundant, and actionable with available data?	Core KPI pack + optional context-specific KPIs
5. Operationalise KPI	Make KPIs usable across contexts within the system	How is each KPI measured, interpreted, and updated over time?	KPI measurement cards (definition, scale)

4 Conceptual framework for KPIs

Building on the synthesis of theory and practice (D3.1), this chapter introduces our four-pillar KPI framework, adapted from the BlueRev conceptual model. We define each category, Environmental, Social Innovation, Governance and Business Model, explain why it matters for method selection and establish a shared vocabulary to ensure that every pilot region evaluates potential practices against the same clear, comparable criteria.

Table 2: Four Pillars KPI framework

Category	Definition
Environmental	Metrics quantifying resource use, waste valorisation, emissions and overall footprint of a method.
Social Innovation	Indicators of stakeholder engagement, inclusivity, and co-creation depth (e.g., Quadruple Helix adoption).
Governance	Measures of regulatory fit, policy coherence and network strength among actors.
Sustainable Business Model	Value proposition, revenue streams, key activities, cost structures, stakeholder relationships and engagements

5 The Key Performance Indicators

One key task within the project was to translate the high-level framework into concrete indicators. To support applicability across regions, each KPI should be in general documented using a short “measurement card” (one per KPI). This makes results auditable and keeps MCDA scoring transparent.

KPI measurement card (example of template).

- KPI name + pillar
- Operational definition (what counts / what does not count)
- Unit / scale (e.g., kg CO_2e per kg product; %, #)
- Method / tool (e.g., LCA, MFA, SROI, survey, network analysis)
- Primary data source(s) (who owns the data; how it’s collected)
- Frequency (baseline, update cycle; note if time-series is required)
- Aggregation rule (per site / per product / per year; mean/median; weighting)
- Quality checks & uncertainty (minimum dataset, missing-data rule, confidence level)
- Interpretation (what direction is “better”; suggested thresholds/targets if available)

For each of the four pillars, we present here a concise set of metrics ranging from greenhouse-gas reduction to stakeholder participation scores, that capture the most critical dimensions of performance. By anchoring each KPI in the realities of side-stream valorisation and stakeholder co-creation, we enable project teams to quantify and compare methods on a like-for-like basis.

5.1 Environmental Footprint KPIs

The Environmental Footprint KPIs take life-cycle thinking and turn it into specific performance metrics, which creates a coherent provision for establishing the true environmental cost of side-stream valorisation in the blue bioeconomy. Each indicator works under the principles of ISO 14040/44-compliant **Life Cycle Assessment (LCA)**, which locates every process choice as working within a cradle-to-gate boundary, and ensures that the associated climate benefits from taking low-value disposal routes and substituting them for residual, value-added routes are provable. Practically, this means that for each kilogram of product made from fishery by-products or seaweed residues, it is possible to investigate the percentage reductions in CO_2-equivalent emissions against a business-as-

usual baseline; this means that a larger reduction score is an indication of successful decarbonisation, and also refers to the European Green Deal's targets for 2030 [1].

In addition to climate considerations, **Material Flow Analysis (MFA)** introduces mass-balance rigor to the assessment of resource circularity. Rather than consider side-streams as uniform and useless waste, MFA takes into account all input flows, co-products, and unavoidable losses to calculate a Material Valorisation Rate, which represents the percentage of raw by-products that have been successfully transformed into bioactive compounds, bioplastics, or feed ingredients. Gaining a valorisation rate above 90 percent can be used as verification, beyond a tagline, to fulfil the EU Circular Economy Action Plan of recovering and reintegrating materials at scale [2].

Energy Intensity encompasses both drivers, resource efficiency and process intensification, by summing the primary energy used, irrespective of its form (steam, electricity, and fuels) and dividing by the final output of the unit operation in kilograms. This KPI, derived from LCA inventory datasets, identifies hotspots in unit operations, for instance, high-pressure extraction or thermal drying, whilst providing a quantitatively based continuous-improvement rationale in consideration of the Energy Efficiency Directive objective to achieve a 32.5 percent reduction in energy-use by 2030 [3]. Ultimately, as its name suggests, teams can use this measure trend of MJ per kg per pilot or technology options to support an onward trajectory to lean, low-carbon processing [4].

Finally, the **Water Footprint** incorporates freshwater stewardship into the valorisation narrative, measuring both blue-water withdrawals and grey-water loads, per kilogram of product. Given that many coastal and island communities are dealing with seasonal water stress, minimizing water use is an important design criterion. Using the LCA water inventory methodology as defined by Hoekstra et al., this indicator not only protects local water security, but also demonstrates contribution to the intended outcomes of the EU Water Framework Directive for sustainable water management [5, 6]. Through standardized LCA + MFA modelling, project teams can thus compare valorisation pathways on a like-for-like basis, find environmental hotspots, and report against climate neutrality, resource circularity, and freshwater stewardship transparently.

5.2 Social Innovation KPIs

In the context of the blue bioeconomy, social innovation is a fundamental driver for inclusive and sustainable transformation. In this deliverable, we approach social innovation not just as a process of stakeholder engagement or as user-

centred design but as a systemic force that can reshape roles, relationships, and governance within value chains. The KPIs developed under this pillar are drawn from literature on co-creation, transformative change, and inclusive innovation and are empirically defined by BlueRev's collective engagement, including multi-actor activities across participating pilot regions. By utilizing metrics such as the Stakeholder Participation Index, Co-Creation Depth Score, and Capacity Building Metrics, projects can effectively measure inclusiveness and the depth of collaboration among diverse actors in the Quadruple Helix model.

The selected KPIs describe various dimensions of social innovation processes:

Stakeholder Participation Index tracks who and how many stakeholders were engaged, mapping participation across four sectors: public, private, academic, and civil society (Quadruple Helix). It emphasizes inclusiveness in innovation ecosystems [7, 8].

Co-Creation Depth Score reflects the level of stakeholder influence in the method-development lifecycle, specifically: if actors were passive consultation only, if they acted as part of active design and prototyping, and if they were part of iterations [9].

Capacity Building Metrics (Number of event and number of people attending) are a reflection of an effort to democratize knowledge and build local agency/competence so that people in the community will adopt the innovation for the long term [10, 11].

By incorporating these evidence-based metrics through the KPI framework, the project provides a baseline for determining whether or not the proposed solutions are really empowering local actors' agency and a vector to build social capital and not simply technical solutions. It is essential to note that these indicators do not presuppose that all participation equates to transformation; rather, they are meant to differentiate between superficial participation and actual co-production of value with a place or community.

In BlueRev, we validated the KPIs through a) stakeholder interviews and b) Living Labs where local actors were able to provide qualitative insights into power dynamics, ownership, and how they build trust. The KPIs were operationalized through participatory tools (focus group, workshops, etc.) with the help of system mapping and Social Return of Investment (SROI) methodologies that contextualize the production of social value beyond strictly economic ways of measuring value.

The SROI framework allows for a holistic approach to measuring the value that initiatives create in terms of social, environmental, and economic outcomes. This

is in contrast to traditional ROI, which typically focuses on the monetary value generated by an initiative. Using SROI, stakeholders can see the whole picture of how and whether initiatives created value by also incorporating intangible benefits. The SROI framework provides a process to involve stakeholders, to assign financial proxies to changes that occur (intense, moderate, or neutral), and to produce a ratio that describes the social value created per euro invested.

The two key components of SROI are that stakeholders are deeply involved and through SROI intangibles can acquire a value.

In order to identify, describe and quantify social impacts SROI engages heavily the target stakeholders. In one study that proposed pathways for managing patients in Spain, the process involved a diverse range of professionals from healthcare and patients themselves [12]. Moreover, SROI helps putting a monetary value on intangible outcomes. For example, the Italian Urban Green Spaces project showed a return of €2 for every euro invested in nature-based solutions [13]. SROI can be applied across multiple sectors such as health, sport, and urban safety, and an SROI study of a football club demonstrated a social value of 2.98:1: important social value created [14]. SROI provides a useful structure for understanding social value, but critics note that the challenge of measuring intangible benefits may overlook the qualitative aspects of social impact. That said, its cross-sector application demonstrates its possibilities for improving the allocation of resources and contributing to community well-being. Multi-Criteria Decision Analysis (MCDA) represents a structured decision-making approach enabling evaluation of options against multiple often conflicting criteria. In contrast to single-criteria approaches (cost-benefit analysis, for example), MCDA may facilitate qualitative and quantitative data to account for stakeholder preferences, and can be used when trade-offs need to be made between three dimensions - economic, environmental, and social - when facing complexity in decision-making. The Analytic Hierarchy Process (AHP) and PROMETHEE are two of the more widely used MCDA methods that promote transparency in prioritising considerations by demonstrating weights assigned to criteria and the scoring of alternatives for a relative measure of merit. In the context of the blue bioeconomy, MCDA is beneficial for decision-makers tasked with designing appropriate "good practice" while balancing the technical feasibilities of the blue bioeconomy, stakeholder social acceptability, and achieving sustainable outcomes [15, 16].

This is particularly relevant to innovations within the blue bioeconomy sector where obtaining and maintaining a social license to operate (if involved in developing a new product, a new market, or a new technology) is as important to success as technical efficiency or market value. It has become fairly well

established that considerations of social equality, equity and justice are central to being effective in our decision-making and maybe arguably we should all collectively strive for some semblance of social acceptance, if not outright representation. Recent studies have suggested that social acceptance can, in some contexts, be more consequential in determining whether or not a project progresses or is deemed a success than even traditional economic considerations or technical parameters [17].

For practitioners operating within an MCDA framework, AHP and PROMETHEE both present structured forms (i.e. mathematical models) that we can use to include these social aspects in a considered and structured way. They allow competing or conflicting priorities to be explicitly motivated and aligned, as well as incorporated to allow stakeholders to represent their views or opinions and reflect accordingly and reconcile different value systems using a quantifiable (or mathematical) [18].

In the marine aquaculture sector, practical examples of these approaches can be seen in relation to Mediterranean finfish and shrimp farming. Their production routinely includes social indicators e.g. stakeholder acceptance, conflicts with other sea users, jobs and equity, alongside bio-physical and cost criteria; sensitivity to social-criteria weights can flip site rankings [19]. Similarly, PROMETHEE has been used with coastal managers to rank defence measures using participatory social weighting [20]. Further, Gomez-López et al., (2008) conducted sensitivity analyses across a variety of MCDA studies and acknowledged that changes in the weight of the social criteria in relation to other criteria may coalesce in dramatic outcomes one way or the other, which supports a different trajectory for the project.

Despite gains in practice, embedding social indicators in MCDA frameworks will still demonstrate notable obstacles. Generally, social dimensions are perceived to be subjective and difficult to measure in relation to, say, environmental emissions or financial returns. It is this subjectivity that contributed to inconsistencies in how the criteria were weighted or prioritized and ultimately influenced decision outputs. When addressing social dimensions, we recommend formats with expert panels, participatory scoring, and scenario-based analyses to be more rigorous and sensitive [21].

Overall, embedding social KPI indicators in MCDA frameworks not only allows for broader and more equal evaluations and assessments, but also to input legitimacy into the innovations introduced to create sustainability as blue bioeconomies.

This collective work contributes to positioning social innovation not as an afterthought but rather a fundamental performance dimension that influences the legitimacy, scalability, and sustainability of good-practice methodologies.

5.3 Governance KPIs

Governance plays a decisive role in shaping the conditions under which blue bioeconomy value-chains can develop and expand. While innovation depends on technological advances, social acceptance, and viable business models, it also requires adaptive and coherent governance frameworks that enable experimentation, foster collaboration, and connect local initiatives with European strategies [22, 23]. The KPIs under this dimension are therefore designed to capture how regulatory processes, policy coherence, institutional networks, and stakeholder perceptions influence the potential for systemic change. By combining both structural and perception-based measures, these indicators make it possible to assess governance not only as a framework for compliance, but as an active driver of transformation in the blue bioeconomy.

In this context, the **Regulatory Adaptation Index** measures the time required for approval of new methods under the European Maritime, Fisheries and Aquaculture Fund (EMFAF) and the Common Fisheries Policy (CFP). This KPI reflects the responsiveness of regulatory systems to innovation. Long approval processes often act as barriers to experimentation, while shorter timelines demonstrate flexibility and an enabling environment for SMEs, start-ups and local actors engaged in the valorisation of side-streams[22].

The **Policy Coherence Score** assesses the extent to which local and regional policies align with overarching EU strategies, particularly the EU Bioeconomy Strategy [24]. Policy fragmentation can dilute the impact of innovative solutions, whereas coherence ensures that regional initiatives reinforce European objectives for sustainability, circularity, and competitiveness [25]. This KPI is therefore indicative of the degree to which governance systems provide consistent and mutually reinforcing directions of change.

Network Density captures the number of formal collaborations within governance bodies in the pilot regions. Dense and resilient networks are crucial for trust-building, knowledge sharing, and collective problem-solving [26]. In practice, this KPI signals whether governance operates through fragmented institutions or through well-coordinated structures capable of mobilising resources and aligning diverse interests.

Beyond structural indicators, two KPIs focus on perceived improvements in understanding: (i) of governance drivers and barriers, and (ii) of the potential for

upscaling blue value-chains. These perception-based measures acknowledge that governance is not only about rules and structures but also about how stakeholders experience and interpret them. Improved understanding strengthens actors' ability to navigate complex institutional landscapes, to identify leverage points for change, and to develop realistic pathways for growth [23].

Finally, the **Number of policy solutions** proposed acts as a proxy for the generative capacity of governance systems. Effective governance is not limited to enforcing compliance; it should foster innovation in policy itself. A higher number of policy solutions emerging from governance bodies demonstrates that actors are actively seeking to remove bottlenecks, design enabling instruments, and contribute to the broader evolution of blue bioeconomy governance [25].

Taken together, these KPIs provide a multidimensional lens on governance performance. They combine hard measures (approval times, policy alignment, collaboration density) with softer but equally important indicators of perception and problem-solving capacity. Within BlueRev, they allow project teams to evaluate governance not as a static backdrop but as an active driver of legitimacy, innovation, and upscaling. In this way, governance KPIs complement the environmental, social, and business model indicators, ensuring that systemic change in the blue bioeconomy is supported by institutional arrangements that are adaptive, coherent, and collaborative.

5.4 Business Model KPIs

The development of sustainable business models in the blue bioeconomy requires a fundamental shift toward circular, regenerative, and inclusive practices. To support this transition, a set of KPIs has been developed, drawing on a combination of methodologies (which were also applied in analyses in previous deliverables -D3.6; D4.2; D4.4-) Responsible Design Thinking (RDT), the Sustainable Business Model Canvas (SBMC), and co-creation workshops. Together, these frameworks provide a robust foundation for stakeholder-driven innovation and systems-oriented sustainable business model development.

Whereby Design Thinking offers a structured, iterative, and human-centred approach to addressing complex challenges. It emphasises creativity, empathy, and user-focused problem solving. Within the Bluerev project, this approach is further enriched by RDT, which integrates ethical considerations and prioritises environmental and social impacts alongside economic viability—creating a more holistic foundation for sustainable innovation. The SBMC complements this by explicitly incorporating ecological and social dimensions into business model design and analysis. It provides a flexible yet systematic framework to align

strategies with circular economy principles, enabling stakeholders to co-develop and refine models that balance environmental sustainability, social value creation, and economic resilience.

Co-creation workshops have been central to this process, engaging a broad spectrum of stakeholders using the Triple Helix Model, which brings together academia, industry, and public sector actors. These participatory sessions ensure that local knowledge, sectoral insights, and governance perspectives are fully integrated into business model innovation.

To ensure these methodologies lead to measurable outcomes, it is essential to apply KPIs that capture the core dimensions of sustainable business models—namely: environmental performance, circular resource use, social value creation, and inclusive stakeholder engagement. The following section presents the most relevant KPIs that have guided the development, assessment, and refinement of sustainable business models developed for the marine and coastal contexts of the BlueRev project.

These KPIs are structured around four core dimensions:

1. Participatory Innovation and Process Effectiveness
2. Circular Resource Utilisation
3. Environmental Impact Reduction
4. Social and Economic Value Creation

Each KPI is described in detail below, including its purpose, relevance, and suggested measurement approach.

Problem–Solution Fit, this KPI is one of the most critical early-stage indicators. It reflects how well the developed business model or solution aligns with the actual needs of intended users—such as SMEs, fishers, small processors, or local communities. A strong fit demonstrates that innovation is grounded in real-world demand, thereby increasing the likelihood of adoption and impact. It ensures that solutions are not only innovative but also contextually appropriate, accepted, and likely to be adopted.

- Measurement:
    - % of users or stakeholders validating the solution as relevant and fit-for-purpose
    - Qualitative feedback from prototype testing sessions

Iteration Frequency, which monitors how often the business model is revised based on stakeholder or prototype feedback. Frequent iteration signals adaptability and learning capacity, both essential in uncertain or emerging domains like marine circularity. It reflects the responsiveness and adaptability of the innovation process—a core principle of Design Thinking.

- Measurement:
  - Number of distinct iterations made from concept to final model
  - Time between feedback received and applied changes

Stakeholder Satisfaction Index assesses the perceived quality and inclusiveness of the design process. It helps to ensure that stakeholder engagement is not only formal but also meaningful and empowering. Measures stakeholder perceptions of the co-creation and design process, including inclusiveness, transparency, and usefulness. High satisfaction indicates effective engagement and strengthens trust and cooperation among actors in the Triple Helix model.

- Measurement:
  - Post-workshop survey results (e.g., Likert scale 1–10)
  - Net Promoter Score (NPS) adapted to stakeholder experience

Time to Prototype KPI, captures how quickly an idea progresses from concept to a testable form. Shorter timeframes indicate an agile and responsive innovation process, enabling faster feedback loops and early identification of design flaws or opportunities. Delays may highlight coordination or knowledge gaps. This implies the average time required to move from ideation to the creation of a first tangible business model or product/service prototype.

- Measurement:
  - Number of days from ideation workshop to first prototype presentation or test

Together, these KPIs measure the quality and responsiveness of innovation, ensuring it remains user-oriented, dynamic, and reflective of stakeholder inputs.

Circular Resource Utilisation, measures the extent to which by-products, side-streams, or organic waste are valorised or reintegrated into the value chain. This reflects one of the central tenets of the circular economy—keeping materials in use for as long as possible and minimising waste. Core to circular economy

thinking—this KPI captures resource efficiency and innovation in biomass valorisation.

- Measurement:
  - % of total input materials derived from circular or secondary sources
  - Tonnes of side-stream materials reused or upcycled annually

Environmental Footprint Reduction, is another critical KPI, capturing the decrease in carbon emissions, water use, energy consumption, or pressure on marine ecosystems as a result of the business model. This KPI provides evidence of environmental performance improvement and directly supports the climate and biodiversity objectives of the EU. It captures ecological benefits of the business model and aligns with EU Green Deal and Blue Growth targets.

- Measurement:
  - CO_{2e} saved per unit of product
  - % reduction in water or energy use
  - Biodiversity or habitat preservation scores (where applicable)

Sustainable Revenue Share Indicator, tracks the proportion of revenue generated through circular, eco-labelled, side-stream based or low-impact products and services. This not only reflects the market viability of sustainability but also demonstrates business alignment with evolving consumer values and regulatory trends. It indicates financial viability and market acceptance of circular business models.

- Measurement:
  - % of total annual revenue from sustainable offerings
  - Growth rate of sustainable revenue streams

Community Value Contribution KPI, which includes metrics such as local employment generation, capacity-building efforts, and reinvestment into coastal communities. This aligns with Bluerev's emphasis on inclusive, place-based innovation. It is essential for inclusive growth, especially in regions reliant on marine resources for livelihoods.

- Measurement:
  - Number of local jobs created (direct and indirect)
  - % of profit or investment returned to local actors
  - Level of capacity-building or training offered to local stakeholders

These KPIs together enable a comprehensive evaluation of the economic, environmental, and social sustainability of each business model under development, offering a structured pathway for improvement and strategic alignment.

The Stakeholder Diversity Index is used to assess the breadth of stakeholder engagement across sectors and roles, ensuring that voices from different domains and demographics are represented. This enhances legitimacy, system insight, and shared ownership of the business model development. It reflects alignment with co-creation principles and strengthens legitimacy and knowledge diversity.

- Measurement:
  - Number of stakeholder groups represented (e.g., industry, academia, civil society, government)
  - Geographic and demographic diversity of participants

Idea Adoption Rate tracks how many concepts or proposals generated during co-creation workshops are carried forward into actual pilot designs or embedded into final business models. It reflects the practical utility and influence of the co-creation process as a tool for generating actionable outcomes.

- Measurement:
  - % of co-created ideas integrated into prototypes, business models, or pilot projects
  - Number of workshop-generated concepts implemented within 6 months

Actionable Outcomes KPI measures the number of concrete initiatives—such as prototypes, partnerships, or policy engagements—that emerge as direct results of co-creation. This underscores the tangible value of participatory innovation.

- Measurement:
  - number of follow-up initiatives within 6 months

Lastly, Co-Creation Satisfaction gauges the perceived effectiveness, transparency, and usefulness of the engagement process from the participants' perspective. High satisfaction levels indicate well-facilitated workshops that balance consultation with action.

- Measurement:
  - average feedback score (1–10 scale)

These KPIs ensure that stakeholder engagement is meaningful, transformative, and well-integrated into the overall business model development process. They help to avoid tokenistic participation and promote long-term collaboration and ecosystem-building.

6 Tools for selecting good practice methods

Selecting the right indicators is only half of the challenge; the other half is choosing the analytical and participatory methods that will generate reliable, actionable data against those KPIs. In this chapter, we introduce a toolkit of approaches from one-on-one interviews and focus groups, system-mapping and business-model canvases, rigorous Life Cycle and Material Flow Analyses, digital traceability solutions, to multi-criteria decision techniques. For each tool type, we explain its purpose, typical steps, and the outputs it produces, so that teams can tailor their approach to the data they have and the decisions they have to make.

6.1.1 Participatory Tools

Purpose: Capture stakeholder insights, build ownership of selected methods, and surface local barriers/enablers.

- **Semi-Structured Interviews** [27, 28]
  - *When:* Early scoping to uncover tacit knowledge, for example in regard to method feasibility, understanding and perception.
  - *How:* 1–1 interview (30–60 min) guided by a flexible protocol (see D3.2) to probe perceived risks, resource needs and social acceptability.
  - *Output:* Thematic codes and a preliminary “feasibility score” for each candidate method.
- **Focus Groups** [29, 30]
  - *When:* To validate and enrich individual interview findings across diverse actors in the value chain.
  - *How:* Facilitated discussions around method performance against draft KPIs; use live polling or scoring to rapidly converge on preferred options.
  - *Output:* Consensus rankings and verbatim insights for qualitative KPI calibration.
- **Workshops & Living Labs** [31, 32]
  - *When:* To co-design, prototype and iteratively test methods under real-world conditions.
  - *How:* Multi-day sessions where stakeholders map out process flows, run small-scale pilots, then reflect on environmental and social outcomes.
  - *Output:* Refined method protocols, refined KPI thresholds, and a “road-test” assessment of operational challenges.

6.1.2 Analytical Frameworks

Purpose: Structure complex information, reveal feedback and align methods with system leverage points.

- **System mapping** [33, 34]
  - *When:* At the very beginning, to make explicit the dynamic interactions among environmental flows, governance rules and social behaviours.
  - *How:* Map variables (e.g., by-product generation, regulatory stringency) and link them with reinforcing or balancing loops; identify where interventions will yield the greatest leverage. .
  - *Output:* Visual loop diagrams highlighting potential “lock-ins” and key feedback to break or reinforce.
- **Sustainable Business Model Canvas (SBMC)** [35]
  - *When:* To assess the viability of circular-economy or bioeconomic business models alongside social and environmental dimensions.
  - *How:* Adapt the nine-block Canvas by adding dedicated sections for resource flows, stakeholder value propositions, and eco-impact; score each block on a 1–5 sustainability scale.
  - *Output:* A tabulated SBMC report showing strengths/weaknesses and guiding where to improve method design.
- **Governance & Innovation Systems Analysis (TIS/GOIS)** [36, 37]
  - *When:* To evaluate enabling conditions—policy instruments, institutional networks, and finance flows—that support method uptake.
  - *How:* Map actors, rules, and resources into an innovation-system framework; assess system functions (e.g., knowledge diffusion, market formation) against KPIs.
  - *Output:* A heat-map of governance strengths and gaps, informing where policy adjustments or capacity building are needed.

6.1.3 Quantitative Assessment Tools

Purpose: Provide rigorous, numerical evaluations of environmental and social impacts.

- **Life Cycle Assessment (LCA)** [38]
  - *Steps:* (1) Goal & scope definition (select KPI targets), (2) Inventory analysis, (3) Impact assessment (e.g., GWP, water use), (4) Interpretation.
  - *Use:* Compare candidate methods on a like-for-like basis, flagging those that exceed footprint thresholds.

- *Output:* Comparative impact profiles (e.g., kg CO_2-eq per kg product).

- **Material Flow Analysis (MFA)** [2]
  - *Steps:* Chart all material inputs and outputs; quantify side-stream valorisation potential.
  - *Use:* Identify methods that maximize conversion of biological residues into value-added products.
  - *Output:* Sankey diagrams or mass-balance tables.
- **Social Return on Investment (SROI)** [39]
  - *Steps:* Engage stakeholders to value social outcomes (e.g., jobs created, equity gains), monetize them, then compute SROI ratio.
  - *Use:* Prioritise methods that deliver the highest societal benefit per euro invested.

6.1.4 Digital & Traceability Tools

Purpose: Enhance transparency, provenance tracking and spatial analysis of resource flows.

- **Blockchain-Based Traceability** [40, 41]
  - *How:* Implement a permissioned ledger to record each step of biomass collection, processing and distribution.
  - *Benefit:* Immutable audit trails build trust among partners and end-users.
- **Geographic Information Systems (GIS)** [42]
  - *How:* Overlay resource supply points, processing facilities and end markets.
  - *Benefit:* Spatial visualisation of hotspots for intervention and optimisation of logistics.

6.1.5 Multi-Criteria Decision Analysis (MCDA)

Purpose: Integrate diverse KPI domains into a single, transparent decision framework.

- **Analytic Hierarchy Process (AHP)** [43]
  - *How:* Stakeholders perform pairwise comparisons of criteria (e.g., environmental vs. social) to derive weights, then score each method accordingly.
  - *Output:* A ranked list of methods with sensitivity analysis on criterion weights.
- **PROMETHEE & ELECTRE** [15, 44, 45]

- *How:* Use outranking techniques when criteria are non-compensatory; define preference functions for each KPI.
- *Output:* Preference flows indicating the most robust good-practice methods.

7 Tool-Selection Guidance

Choosing the right combination of tools is crucial to ensure that KPI measurement and method selection are both efficient and fit-for-purpose. This section offers practical guidance on matching tools to your specific project conditions. We first consider the nature of available data, whether quantitative datasets or qualitative insights, then assess stakeholder capacity and institutional readiness to apply various approaches. Finally, we address the broader decision context, highlighting which tools best support economic, social or regulatory priorities. By following this guidance, project teams can design a streamlined, context-sensitive evaluation workflow that leverages the strengths of each tool while avoiding unnecessary complexity.

- **Data Availability:**
  - If primary data on emissions or material flows exist, prioritise LCA/MFA.
  - If only qualitative insights are available, lean on participatory tools and CLDs.
- **Stakeholder Capacity:**
  - For regions with strong local governance networks, TIS analysis and living labs enhance co-design.
  - Where decision makers demand clear rankings, MCDA approaches (AHP, PROMETHEE) deliver transparent scores.
- **Decision Context:**
  - Use SBMC when evaluating business model innovation; combine with SROI for social ventures.
  - Embed blockchain or GIS when traceability or spatial planning is a regulatory or market requirement.

By combining these tools—in phased sequences or parallel tracks—project teams can robustly screen, evaluate and select the good-practice methods that best balance environmental footprint reduction with social, governance and economic viability.

8 Applying KPIs & Tools in Pilot Regions

Translating a generic KPI framework and toolkit into actionable regional strategies requires tailoring indicators and methods to local realities. In this chapter, we illustrate how Denmark/Greenland, Estonia and Italy can deploy the metrics and tools outlined above to their unique side-stream valorisation challenges. Each vignette demonstrates which KPIs are most relevant, how to gather and interpret data using the selected tools, and how to integrate findings into decision-making processes. These examples serve as practical blueprints, guiding future teams through the steps of local adaptation, data collection and method validation in diverse bioeconomy contexts. While detailed results will appear in different project outputs, illustrative applications include:

- **Denmark:** Transforming fish by-products (salmon) into premium Omega-3 health drinks, combining sustainability and health benefits (for export markets). Specific case: Omega-3 Functional Beverage
- **Greenland:** Valorising fish by-products integrating circular practices and waste management. Specific case: Dog food from fish processing waste.
- **Estonia:** Tracking new algae-based product adoption (% market uptake) and mapping stakeholder network density in governance forums.
- **Italy:** Marine By-Product and side-streams Transformation and Valorisation. Specific case: Ritunnu – Sustainable valorisation of underutilized(by-catch) low-value fishery species

9 Discussion and recommendations

In order to maximize the disruptive potential of social innovation in the context of the BlueRev project, and more broadly in the context of the blue bioeconomy, some vital disposition are brought to our attention as a result of this experience, and each of these points encourages us to think more deeply about what it means to move beyond traditional stakeholder engagement to fully activated social processes that solidifies legitimacy, inclusivity, and sustainability in the world of innovation.

It is important, first and foremost, that social innovation indicators are considered as part of an evaluation framework and not as peripheral or “soft” indicators. In practice, this means social Key Performance Indicators (KPIs) need to be mainstreamed into the formal assessment process - such as Multi-Criteria Decision Analysis (MCDA) - so that they carry equal weight as environmental or technical indicators. By embedding social KPIs into the connection with these evaluation protocols, projects are able to ensure that community value, equity, and social cohesion are no longer on the periphery, but rather critical to describing and evaluating how success is defined.

Also, stakeholder engagement has to move beyond participation. Sadly, stakeholder engagement often only amounts to consultations that are part of a decision-making process that is taking place elsewhere. Real social innovation involves moving to co-ownership, so that stakeholders are not simply the subjects of the work, but co-designers and co-decision-makers of the strategies/interventions. The level of co-creation has to be assessed by more than quantifiable metrics; qualitative evidence, including stories about how the actors brought about actions and/or took leadership roles, to provide much-needed context and to avoid being too superficial when assessing the quality of engagement.

Another important aspect is the use of Social Return on Investment (SROI) approaches so that the social value created by an intervention is not only captured but also made quantifiable. The social impacts of interventions, such as jobs created, social inclusion by building trust, or being aligned with equity, can often remain untracked, unacknowledged, or otherwise not taken into consideration in standard evaluations. In certain contexts, SROI takes a progressive step forward by providing a framework that represents these impacts in financial terms. SROI can build capacity to inform investment proposals, justify expenditures of public money, and support evidence-based policymaking and practices associated with social welfare, as dictated by the societal well-being ethos.

It is equally important to understand that social innovation is not a linear process that can be replicated identically across contexts. Both the KPIs used to assess social progress and the engagement process with stakeholders must be culturally and contextually flexible and adaptable. For example, what inclusive governance means can differ greatly in the context of various regions. Concepts of trusted leadership, collaborative scenarios, and participatory systems may be structured in some communities in northern Greenland, one way, while they are shaped completely differently in a Sicilian fishery cooperative or other regional contexts. The need to locally calibrate indicators and the types of means for engagement is vital if the context is going to have any relevance, legitimacy, and ultimately, be effective.

In the end, there need to be commitments to investing in the processes that enable meaningful social innovation. The character of co-creation (as assessed through indicators or otherwise) and the quality of participation (can vary) is directly contingent upon the identity of the skilled facilitators. The social innovation process has a myriad of challenges (such as language, power inequalities, and/or competing interests) that need to be mediated with specific process supports in place to actively address these challenges (i.e., whether used Living Lab coordinators or trusted community intermediaries, or a specific skilled facilitator to facilitate dialogue for example to build trust and develop shared normative organizational knowledge). The types of process supports should be conducted such that stakeholders feel their voice is heard and respected. Without some form of institutional support for social innovation to scaffold upon, social innovation is likely to remain aspirational rather than actual.

In conclusion, social innovation in blue bioeconomy projects like BlueRev requires intentional efforts to develop strategies that promote genuine ownership, transparency, and impact, in addition to inclusion. The considerations that have been presented here represent starting points for the design of interventions that are effective, equitable and socially resilient.

10 Conclusions

This deliverable (D3.3) turns BlueRev's conceptual insights (D3.1) and field evidence (D3.2) into a practical decision architecture that pilots can use to identify, compare and justify "good-practice" methods in the blue bioeconomy. The work provides a four-pillar KPI framework, covering environmental performance, social innovation, governance and business models, together with a toolkit of participatory, analytical, digital and MCDA methods and clear guidance on how and when to apply them. Assessment therefore moves beyond narrow footprint metrics to an integrated view that combines technical performance, social legitimacy, policy fit and economic viability.

A principal conclusion is that the framework establishes a common language for evaluation. Pilots and partners can now compare options on a like-for-like basis while retaining the flexibility to tailor indicators and thresholds to local realities. On the environmental side, the combination of ISO-compliant LCA, MFA-based valorisation rates, energy intensity and water footprint provide the necessary granularity to detect process hotspots, quantify circularity outcomes and connect results directly to EU climate, circular-economy and water objectives.

Equally important, social innovation emerges as a true performance domain rather than a peripheral "soft" consideration. Stakeholder Participation, Co-creation Depth, Capacity-building measures and SROI make the social dimension measurable and auditable. Embedding these criteria within AHP or PROMETHEE gives social licence, equity and community value the same decision weight as technical and cost parameters. Sensitivity analyses show that changing the weight of social criteria can alter the preferred option, which means social metrics must be treated as decision-critical rather than ornamental.

Governance conditions are shown to be decisive for scale. Indicators such as Regulatory Adaptation, Policy Coherence and Network Density, combined with perception-based measures of barriers, drivers and the generation of policy solutions, offer a workable way to judge whether institutions are ready to host and expand new methods. In parallel, business-model viability is framed as integral to sustainability.

Responsible Design Thinking and the Sustainable Business Model Canvas are accompanied by KPIs for problem-solution fit, iteration, time-to-prototype, circular resource use, footprint reduction, sustainable revenue share, community benefits, stakeholder diversity, adoption of ideas, and change-actionable outcomes. These KPIs can be used as evidence base during the earlier stages

of the co-creation processes and allow for pivots and change of directions of the desired transitions as knowledge increases during the process.

Assistance with tool selection diminishes ambiguity by indicating methods matched to operational capacity from the data-reliant, community-engaged, and shared decision production facets of the report. In practice teams can map webs of interconnected truths, problematise with participatory diagnosis, quantify impacts with LCA, MFA and SROI, and synthesise the results with MCDA for traceable ranking of implications on performance. Outputs go to the WP 5 dashboard and into WP6 Policy Labs, providing a traceable performance narrative through analysis, deliberation and action.

Early applications in Denmark/Greenland, Estonia and Italy indicate that the framework is transferable across diverse product pathways, from fish by-products destined for pet food or health ingredients to algae innovations, while preserving comparability and auditability. Open-science practices and a robust data-management plan ensure FAIR data across work packages, and digital traceability tools such as GIS and blockchain strengthen provenance and routine KPI reporting.

For BlueRev, these advances change how choices are made and defended. Project selection becomes more transparent because shortlisted methods must meet agreed thresholds across all four pillars, not only minimise emissions. Learning loops are built into the process: KPI trends and MCDA results can be revisited after Lighthouse feedback and Policy Labs, which operationalise adaptive governance. Policy dialogue is also better equipped; governance KPIs and SROI provide concrete evidence for regional briefs and EU-level discussions on permitting, procurement and traceability.

Some limitations remain. Data heterogeneity, especially for social metrics and uneven institutional capacity, can affect comparability across regions. Standardised protocols, minimum data packages and explicit uncertainty ranges will mitigate these issues, but continued attention is required. Several business-model indicators depend on time series and should be tracked.

To support consistent and transparent performance assessment, the BlueRev KPI framework is summarised as a structured, step-by-step procedure guiding users from the definition of the decision context to the interpretation of KPI-based results. By explicitly linking sustainability objectives to indicator selection, operationalisation, and assessment, the approach enables comparable and evidence-based evaluation of blue bioeconomy valorisation pathways across different contexts. This clear procedural logic strengthens the practical applicability of the framework and provides users with concrete guidance for building and assessing tailored KPI sets.

In conclusion, Deliverable D3.3 highlights how BlueRev pilots not only tested local solutions but also generated transferable knowledge that can guide Europe's coastal regions in their transition. By connecting governance, innovation, and community resilience, the project offers a practical pathway towards a more inclusive and sustainable blue bioeconomy.

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